Type MS · Writing down the asymptotes of a rational curve

Answer 1 of 19 · Question 3 part d · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number		
3 (a)	$c=2,\ d=3$	B1B1 [2]
(b)	$a-bx=0 \Rightarrow x=\frac{a}{b}=\frac{5}{4} \Rightarrow a=5,\ b=4$	M1A1 [2]
(c)	When the curve crosses the y-axis $x=0$ $y=\frac{5-4\times0}{2\times0-3}=-\frac{5}{3} \Rightarrow p=-\frac{5}{3}$	M1A1 [2]
(d)	$y=-2$	B1ft [1]
		Total 7 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 1 of 19 · Question 3 part d · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 2 of 19 · Question 3 part b · 1 mark

2023 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

3 a	$\frac{a}{-2}=4\Rightarrow a=-8$	M1 A1 (2)
b	$x=\frac{1}{2}$	B1 (1)
c (i)	$\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$	B1 ft
c (ii)	$(0,3)$	B1 (2)
d	Asymptote drawn at $y=4$. The asymptote must either be clearly labelled with its equation or pass through the y axis with 4 clearly labelled.	B1
	Asymptote with equation drawn at $x=\frac{1}{2}$ The asymptote must either be clearly labelled with its equation or pass through the x axis with $\frac{1}{2}$ labelled.	B1 ft
	Correct curve drawn with two branches	B1 ft
	"$\frac{3}{8}$" and "3" labelled on x and y axes respectively	B1 ft (4)
		Total 9 marks

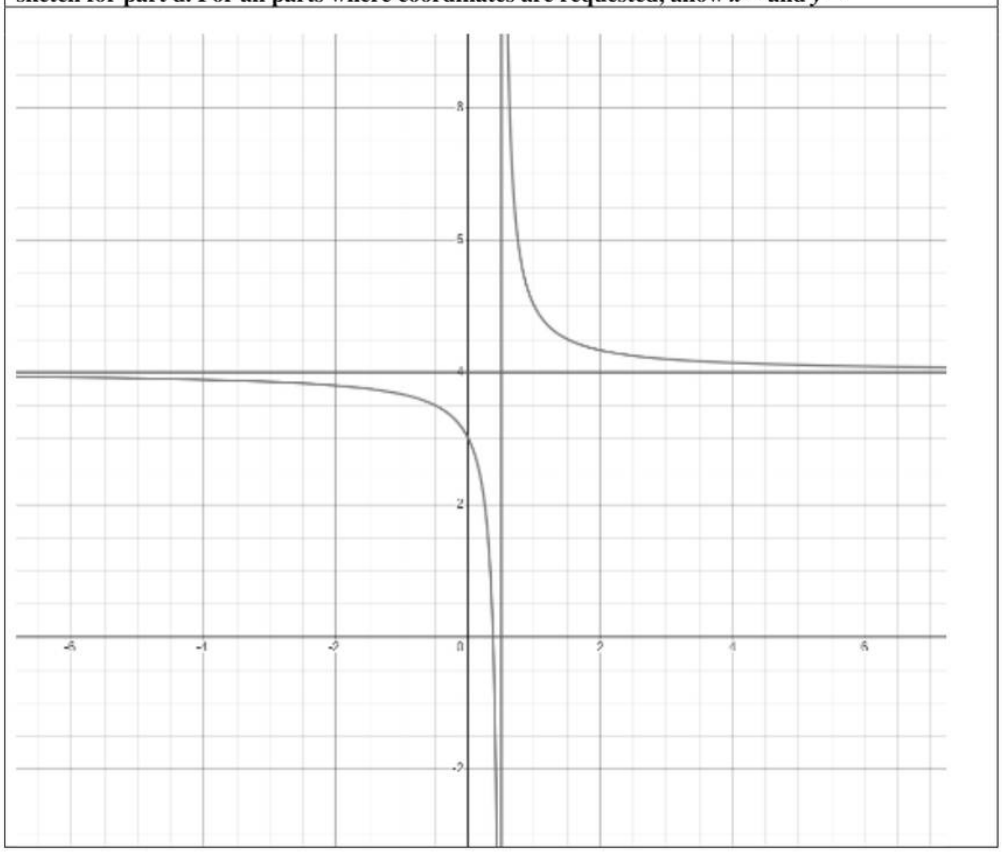
Part	Mark	Additional Guidance
(a)	**M1**	For $\frac{a}{-2}=4$
	A1	For $a=-8$
	Note: it is acceptable if a candidate uses a suitably large value for x and deduces the correct value for a. M1 A1	
(b)	**B1**	For $x=\frac{1}{2}$
(c)(i)	**B1 ft**	For $\left(\frac{3}{-"-8"},0\right)\left(=\left(\frac{3}{8},0\right)\right)$ ft their value from (a). Allow omission of brackets. This mark is awarded for the point at which you see $\left(\frac{3}{-\text{their } a},0\right)$ May be written $x=$
(ii)	**B1**	For $(0,3)$. Allow omission of brackets. May be written $y=$
If there is no labelling of (i) and (ii), it must be unambiguously indicated which coordinate crosses the y axis and which crosses the x axis. If any doubt, do not award the marks.		
(d)	**B1**	For asymptote with equation drawn at $y=4$. The line must be labelled with the equation or must clearly pass through the y-axis labelled as 4. There must be at least one branch of the curve present, which must not cross or bend back from the asymptote.
	B1 ft	For asymptote with equation drawn at $x=\frac{1}{2}$, ft their answer from part b The line must be labelled with the equation or must clearly pass through the x-axis labelled as 0.5. There must be at least one branch of the curve drawn, which must not cross or bend back from the asymptote.

Type MS · Writing down the asymptotes of a rational curve

Answer 2 of 19 · Question 3 part b · 1 mark

2023 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

	B1 ft	For correct curve drawn with two branches, in the correct quadrants, ft their answers from part a and b. The curve must be drawn in quadrants 1 and 3.
	B1 ft	For a curve passing through "$\frac{3}{8}$" and "3" labelled on x and y axes respectively, ft their answers from part a, b and c.
Note: answers for parts a, b and c cannot be awarded marks for correct values labelled on the sketch for part d. For all parts where coordinates are requested, allow $x =$ and $y =$		



Type MS · Writing down the asymptotes of a rational curve

Answer 2 of 19 · Question 3 part b · 1 mark

2023 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 3 of 19 · Question 10 part a.i · 0 marks

2022 JAN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 4 of 19 · Question 10 part a.ii · 2 marks

2022 JAN · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 5 of 19 · Question 10 part a.i · 0 marks

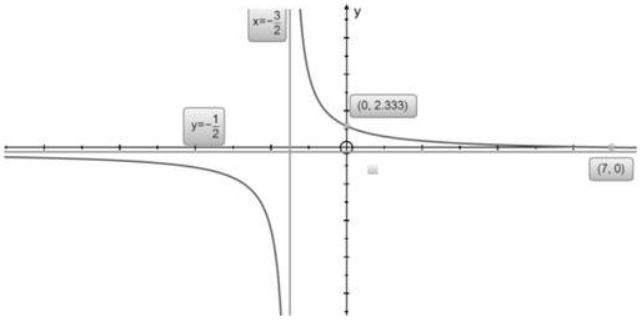
2025 JUN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 6 of 19 · Question 10 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part a.ii in this mark scheme extract

10 (a)	(i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$	B1 B1 [2]
(b)	$(7,0)$ $\left(0,\frac{7}{3}\right)$	B1 B1 [2]
(c)		B1ft curve B1ft asymptotes B1ft (intersections with x- and y-axes) [3]
(d)	$\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$ $\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$ ALT – product rule $-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$ $-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$	M1 M1A1* cso [3] M1 M1A1* cso

Type MS · Writing down the asymptotes of a rational curve

Answer 6 of 19 · Question 10 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part a.ii in this mark scheme extract

(e)	$-\frac{17}{(2x+3)^2}=-\frac{17}{16}$	M1
	$(2x+3)^2=16\Rightarrow 2x+3=\pm\sqrt{16}\Rightarrow x=0.5 \text{ or } -3.5$	dM1
	$y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$	ddM1
	$\left(-\frac{7}{2},-\frac{21}{8}\right)$	A1
	$y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$	M1A1
	$34x+32y+203=0$	A1 [7]
		Total 17 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 6 of 19 · Question 10 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part a.ii in this mark scheme extract

Part	Mark	Additional Guidance
For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.		
(a)	(i) B1	For $x=-\frac{3}{2}$ oe
	(ii) B1	For $y=-\frac{1}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given
(c)	B1ft	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(d)	M1	Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$. Denominator must be of the form $(2x+3)^2$
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants c, d.
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
(e)	M1	For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d)
	dM1	For rearranging to obtain $(2x+3)^2=16$ and attempt to solve or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve.

Type MS · Writing down the asymptotes of a rational curve

Answer 6 of 19 · Question 10 part a.ii · 2 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part a.ii in this mark scheme extract

	ddM1	For substitution of their x into the original equation to obtain y. This must **not** be $x=\frac{1}{2}$ This M mark is dependent on the **two** previous M marks
	A1	For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$
	M1	For substitution of their coordinates of P and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method. If $y=mx+c$ is used a value of c must be found **and** an equation needs to be formed. For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [
	A1	For any correct equation of the tangent with the correct values of x and y in any form.
	A1	For the correct equation of tangent given in the required form. $34x+32y+203=0$

Type MS · Writing down the asymptotes of a rational curve

Answer 6 of 19 · Question 10 part a.ii · 2 marks

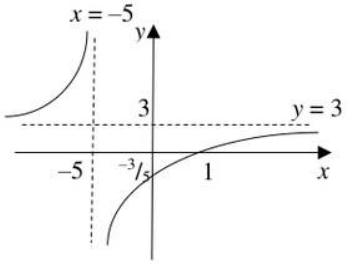
2025 JUN · Paper 02 · extract 5 of 5 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 7 of 19 · Question 3 part b.i · 0 marks

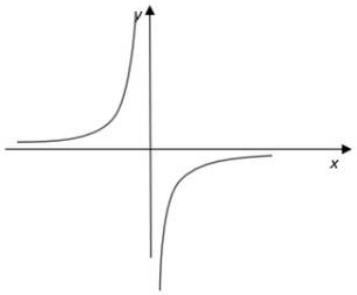
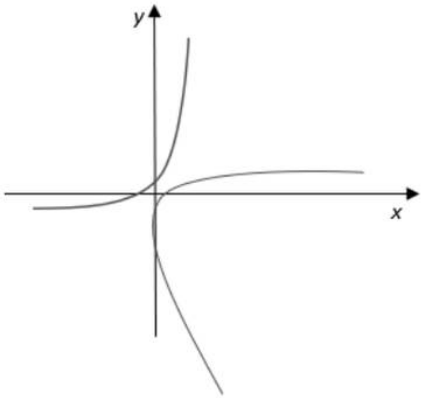
2021 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

3 (a)	$\frac{dy}{dx} = \frac{a(x+5)-(ax-3)}{(x+5)^2}$	M1
	When $x = 2$ $\frac{dy}{dx} = \frac{7a-2a+3}{49} = \frac{18}{49} \Rightarrow a = ...$	M1
	$a = 3$ *	A1cso (3)
3 (b)(i)	$y = 3$	B1
(b)(ii)	$x = -5$	B1 (2)
(c)(i)	$(1, 0)$	B1
(c)(ii)	$\left(0, \frac{-3}{5}\right)$	B1 (2)
(d)	 Curve drawn Asymptotes drawn and labelled $^{-3}/_{5}$, 1 labelled on axes	B1 B1ft B1ft (3)
		[10]
		Total 10 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 7 of 19 · Question 3 part b.i · 0 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

<table>
<tr><td></td><td></td><td></td></tr>
<tr><td>(b)(i)</td><td>B1</td><td>For $y = 3$ This must be an equation of a line. Do not award for just 3.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $x = -5$ This must be the equation of a line. Do not award for just –5</td></tr>
<tr><td>(c)(i)</td><td>B1</td><td>For (1, 0) or clearly listing $x = 1$, $y = 0$ as a pair.</td></tr>
<tr><td>(ii)</td><td>B1</td><td>For $\left(0, -\frac{3}{5}\right)$ oe. or clearly listing $x = 0$, $y = -\frac{3}{5}$ as a pair.</td></tr>
<tr><td rowspan="4">(d)</td><td rowspan="3">B1</td><td>For the curve drawn with two branches anywhere on the grid provided it is a negative reciprocal curve. The ends of the curves must be asymptotic and must not turn back on themselves. Do not allow any obvious overlap across the ends of the curve with evidence of the presence of asymptotes.</td></tr>
<tr><td>For example, accept:
Negative reciprocal curve in incorrect position
 
</td><td>Do not accept:
• Overlap of asymptotes
• Ends turning back on themselves
 
</td></tr>
<tr><td></td></tr>
<tr><td>B1ft</td><td>For their asymptotes correctly drawn and clearly labelled with their equation.
At least one branch of the curve is required. It must be a negative reciprocal and it must be in the correct position for their asymptotes. The follow through is available for their answers in part b.
If correct asymptotes appear on the sketch, do not award marks retrospective marks in part b.</td></tr>
<tr><td>B1ft</td><td>The curve must be drawn going through their two points of intersection. It must be a negative reciprocal, in the correct position for their intersections and clearly marked on the axes. The follow through is available for their answers in part c. Allow the other branch to even be missing.
If correct coordinates appear on the sketch, do not award marks retrospective marks in part c.</td></tr>
</table>

Type MS · Writing down the asymptotes of a rational curve

Answer 7 of 19 · Question 3 part b.i · 0 marks

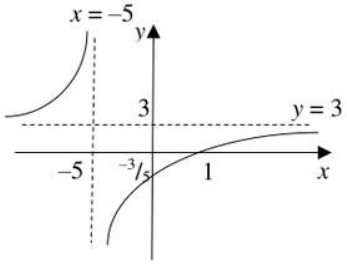
2021 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 8 of 19 · Question 3 part b.ii · 2 marks

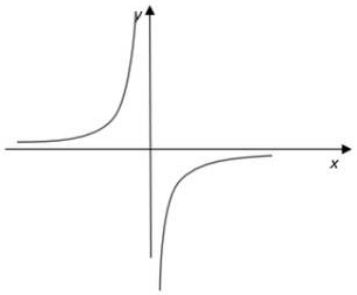
2021 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

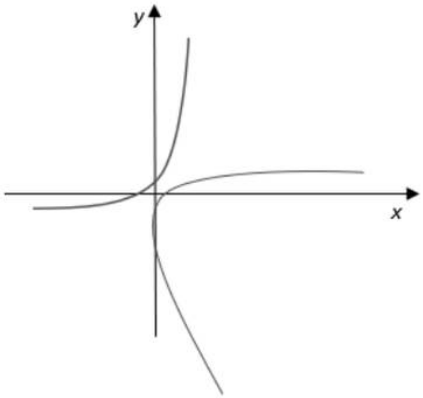
3 (a)	$\frac{dy}{dx} = \frac{a(x+5)-(ax-3)}{(x+5)^2}$	M1
	When $x = 2$ $\frac{dy}{dx} = \frac{7a-2a+3}{49} = \frac{18}{49} \Rightarrow a = ...$	M1
	$a = 3$ *	A1cso (3)
3 (b)(i)	$y = 3$	B1
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(d)	 Curve drawn Asymptotes drawn and labelled $^{-3}/_{5}$, 1 labelled on axes	B1 B1ft B1ft (3)
		[10]
		Total 10 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 8 of 19 · Question 3 part b.ii · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

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<tr><td rowspan="4">(d)</td><td rowspan="2">B1</td><td>For the curve drawn with two branches anywhere on the grid provided it is a negative reciprocal curve. The ends of the curves must be asymptotic and must not turn back on themselves. Do not allow any obvious overlap across the ends of the curve with evidence of the presence of asymptotes.</td></tr>
<tr><td>For example, accept:
Negative reciprocal curve in incorrect position
 

Do not accept:
• Overlap of asymptotes
• Ends turning back on themselves
 
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<tr><td>B1ft</td><td>For their asymptotes correctly drawn and clearly labelled with their equation.
At least one branch of the curve is required. It must be a negative reciprocal and it must be in the correct position for their asymptotes. The follow through is available for their answers in part b.
If correct asymptotes appear on the sketch, do not award marks retrospective marks in part b.</td></tr>
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If correct coordinates appear on the sketch, do not award marks retrospective marks in part c.</td></tr>
</table>

Type MS · Writing down the asymptotes of a rational curve

Answer 8 of 19 · Question 3 part b.ii · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 9 of 19 · Question 7 part a · 1 mark

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

7(a)	$x = -\frac{5}{4}$	B1 [1]
(b)	$\{\frac{dy}{dx}\} = \frac{2x(4x+5) - 4(x^2-1)}{(4x+5)^2}$ $y = 0$ $\{\frac{dy}{dx}\} = \frac{2(-1)(4(-1)+5) - 4((-1)^2-1)}{(4(-1)+5)^2} = -2$ $\rightarrow \{gradient\ of\ normal =\} \frac{1}{2}$ $y - 0 = \frac{1}{2}(x-(-1))oe$	M1A1A1 B1 M1 M1A1cso [7]
(c)	$\frac{x+1}{2} = \frac{x^2-1}{4x+5}$ $2x^2 + 9x + 7 = 0$ ***OR*** $(x+1)(4x+5) = 2(x+1)(x-1)$ $(2x+7)(x+1) = 0 \Rightarrow x = \cdots$ ***OR*** $(4x+5) = 2(x+1) \Rightarrow x = \cdots$ $x = -\frac{7}{2}, \{x = -1\}$ $y = \frac{"-\frac{7}{2}"+1}{2} = \cdots$ $\left(-\frac{7}{2}, -\frac{5}{4}\right)$	M1 dM1 ddM1 A1 M1 A1 [6]
		Total 14 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 9 of 19 · Question 7 part a · 1 mark

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	$x = -\frac{5}{4}$ *or* $x = -1.25$, do not accept 4*x*+5=0
(b)	M1	Attempts to differentiate *y*. **Quotient Rule:** Look for $\frac{Px(4x+5) - Q(x^2-1)}{(4x+5)^2}$, *where* $P > 0, Q > 0$ If the quotient rule is quoted, it must be correct. Condone invisible bracket for this mark. **Product Rule:** Look for $Px(4x+5)^{-1} \pm Q(x^2-1)(4x+5)^{-2}$, *where* $P > 0, Q > 0$
	A1	Quotient rule: Correct denominator $(4x+5)^2$and one correct term in the numerator, 2*x*(4*x*+5) or $-4(x^2-1)$. Product rule: One term correct
	A1	Fully correct. $\left\{\frac{dy}{dx} =\right\} \frac{2x(4x+5)-4(x^2-1)}{(4x+5)^2}$ or $\left\{\frac{dy}{dx} =\right\} 2x(4x+5)^{-1} - 4(x^2-1)(4x+5)^{-2}$oe
	B1	*y* = 0 seen or implied
	M1	Correct method finding gradient of normal to the curve where *x*= -1
	M1	Correct method of forming the equation of a straight line using the point (**-1,their y value**) with a "**changed**" gradient.("changed" means a different gradient from their tangent gradient, their normal could be found by an incorrect method) If using *y*=*mx*+*c*, a value for *c* must be found and the equation formed.
	A1cso	For the correct line in any form, simplified or unsimplified. (isw) Once correct equation is seen, award the mark, ignore later incorrect simplification. e.g.2*y* = *x*+1 or 2*y*-*x*-1=0 or *y*=0.5x+0.5 oe or even $y - 0 = \frac{1}{2}(x - (-1))$
(c)	M1	For equating the equation of their line with the equation of the curve.
	dM1	For forming an equation with all like terms collected. **OR** cross multiplies then factorises e.g. $2(x^2 - 1) = 2(x+1)(x-1)$
	ddM1	Solves their equation with a correct method and finds at least one real value of *x* which is not -1, method of solving could be implied by correct answers if calculator used. **OR** cancels (*x*+1) and solves their linear equation.
	A1	$x = -\frac{7}{2}$, $\{x = -1\}$
	M1	For substituting their *x* value (not -1) to either the equation of their normal or the equation of the given curve to find a value for *y*
	A1	For the correct exact coordinates of $D\left(-\frac{7}{2}, -\frac{5}{4}\right)$ *oe* Allow *x* = -3.5, y = -1.25 instead of given as coordinates.

Type MS · Writing down the asymptotes of a rational curve

Answer 9 of 19 · Question 7 part a · 1 mark

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 10 of 19 · Question 3 part a.i · 3 marks

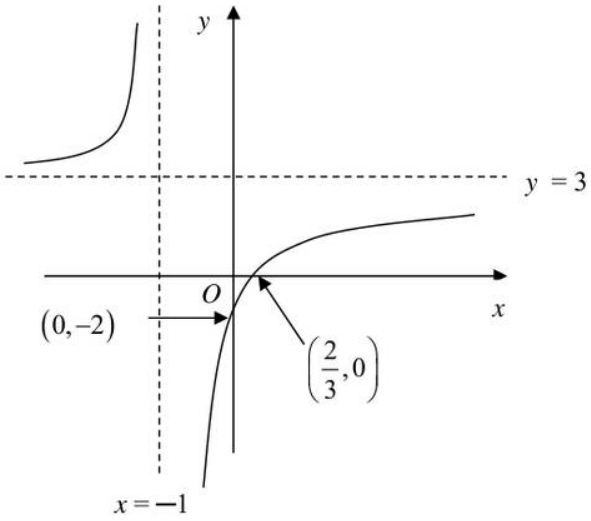
2025 NOV · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 11 of 19 · Question 11 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · extract 1 of 5 · find part a.i in this mark scheme extract

11 (a)	(i) $y=3$ (ii) $x=-1$	B1B1 [2]
(b) (i) (ii)	$0=\frac{3x-2}{x+1}\Rightarrow 3x=2\Rightarrow x=\frac{2}{3}$ $y=\frac{3\times 0-2}{0+1}=-2$	B1 B1 [2]
(c)		B1 (shape and position) B1 (asymptotes) B1 (intersections) [3]
(d)	$mx+4=\frac{3x-2}{x+1}\Rightarrow mx^2+x(m+1)+6=0$ $b^2-4ac<0\Rightarrow (m+1)^2-4\times m\times 6<0\Rightarrow m^2-22m+1<0$ $(m-11)^2-120=0\Rightarrow m=11\pm\sqrt{120}$ Defines region $11-2\sqrt{30}<m<11+2\sqrt{30}$	M1 M1 M1A1 M1 depM1A1 [7]
		Total 14 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 11 of 19 · Question 11 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · extract 2 of 5 · find part a.i in this mark scheme extract

(a) (i)	
B1	$y = 3$
(a) (ii)	
B1	$x = -1$
(b) (i)	
B1	$x = \frac{2}{3}$
(b) (ii)	
B1	$y = -2$
(c)	
B1	Two curves in the correct quadrant (Condone poor shape as long as intention is clear)
B1	Two asymptotes drawn and correctly labelled (Allow –1 and 3 correctly labelled on axes to count as labelled). There must be at least one part of the curve drawn
B1	Intersections labelled
(d)	
M1	Equates C and l (Condone use of $\neq$ and $<$)
M1	Forms a 3TQ (Condone the use of $\neq$ and $<$) (allow 1 error in the expansion of correct brackets or 1 error in simplifying)
M1	Use of $b^2 - 4ac < 0$ to form a 3TQ
A1	Simplify to $m^2 - 22m + 1 < 0$
M1	Solving the 3TQ If the quadratic is incorrect then the method for solving must be shown
depM1	Simplifying to the form $a \pm b\sqrt{2}$ Dependant on previous M1
A1	$11 - 2\sqrt{30} < m < 11 + 2\sqrt{30}$ (Allow $a = 11, b = 30$)

Type MS · Writing down the asymptotes of a rational curve

Answer 11 of 19 · Question 11 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · extract 3 of 5 · find part a.i in this mark scheme extract

Type MS · Writing down the asymptotes of a rational curve

Answer 11 of 19 · Question 11 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · extract 4 of 5 · find part a.i in this mark scheme extract

Type MS · Writing down the asymptotes of a rational curve

Answer 11 of 19 · Question 11 part a.i · 0 marks

2020 JAN · Paper 02 (regional R) · extract 5 of 5 · find part a.i in this mark scheme extract

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Type MS · Writing down the asymptotes of a rational curve

Answer 12 of 19 · Question 11 part a.ii · 2 marks

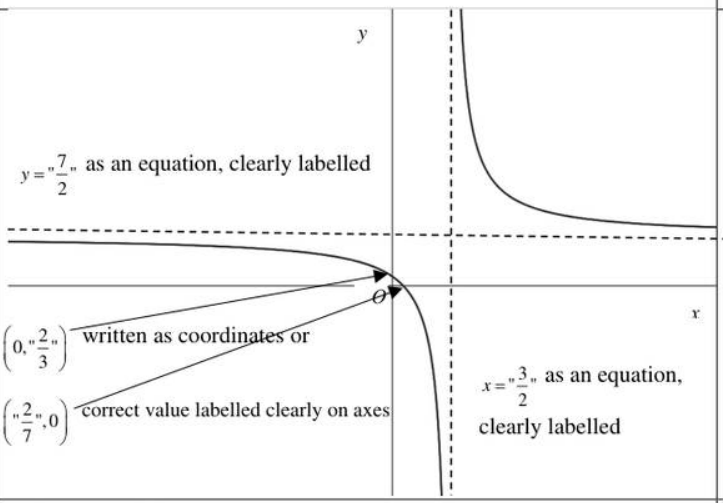
2020 JAN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 13 of 19 · Question 10 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 5 · find part a.i in this mark scheme extract

10 (a) (i)	$x = \frac{3}{2}$	B1
(ii)	$y = \frac{7}{2}$	B1 [2]
(b)	$\left(\frac{2}{7}, 0\right)$ $\left(0, \frac{2}{3}\right)$	B1 B1 [2]
(c)	$\frac{7(2x-3)-2(7x-2)}{(2x-3)^2}$	M1 A1
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1 [4]
	ALT – product rule	M1 A1
	$7(2x-3)^{-1} + (7x-2)(-1)(2)(2x-3)^{-2}$	
	$\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$	A1
	Correct conclusion	B1
(d)		B1 (curve) B1ft (asymptotes) B1ft (intersections with x- and y-axes) [3]
(e)	$-\frac{1}{17} = "\frac{-17}{(2x-3)^2}"$	M1
	"$(2x-3)^2 = 17^2$" **or** "$4x^2 - 12x - 280 = 0$" oe	dM1
	$x = 10$ $y = 4$ or $(10, 4)$	A1
	$y - "4" = 17(x - "10")$ **or** "4" = –17 × "10" + c leading to c =	M1
	$y = 17x - 166$ oe	A1

Type MS · Writing down the asymptotes of a rational curve

Answer 13 of 19 · Question 10 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 5 · find part a.i in this mark scheme extract

	"$17x - 166" = \dfrac{7x-2}{2x-3}$ $\rightarrow$ $17x^2 - 195x + 250$	M1
	$x = \dfrac{25}{17}$ $y = -141$ or $\left(\frac{25}{17}, -141\right)$	A1 [7]
		Total 18 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 13 of 19 · Question 10 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 5 · find part a.i in this mark scheme extract

Part	Mark	Additional Guidance
		If a candidate gives **no response** to (a) and/or (b) but shows the correct answers on the graph we will award the marks. Where answers are given in (a) and/or (b) these should be marked as they stand with no reference to the graph. Ignore labelling of (i) and (ii) and mark (a) together.
(a)(i)	B1	For $x = \frac{3}{2}$ oe
(a)(ii)	B1	For $y = \frac{7}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x = \frac{2}{7}$ and/or $y = \frac{2}{3}$ given
(c)	M1	Attempt the quotient rule. Numerator must be the difference of two terms (either way round) of the form $A.(2x-3) - B.(7x-2)$, A and $B > 1$. Denominator must be of the form $(2x-3)^2$
	A1	Either term on the numerator correct (either way round), dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x-3)^{-1} + d(7x-2)(2x-3)^{-2}$ for constants c, d.
	A1	Either term correct, dependent on previous method mark.
	A1	Obtains $\frac{-17}{(2x-3)^2}$ or $\frac{-17}{4x^2-12x+9}$
	B1	Correct conclusion based on correct working only, for example, (the numerator is a negative number and) the denominator is always positive and therefore the fraction/gradient is always negative.
(d)	B1	Two branches drawn in the correct two "quadrants" created by the two aymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes. Allow BOD if intention is for curve to run alongside asymptote but there is a slight deviation back on itself.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(e)	M1	Sets their differentiated function from part (c) = $-\frac{1}{17}$
	dM1	Rearranges to get to an equation of the form shown with no denominators or a 3TQ and solves using an acceptable method to obtain $x = \ldots$. Dependent on previous method mark
	A1	Correct values for point A (10, 4)
	M1	Uses their values for x and y (from an attempt at working with gradient of the curve) with gradient 17 to find an equation for l (if using $y = mx + c$, must be a complete method arriving at $c =$) If correct $c = -166$.
	A1	Correct equation, any form

Type MS · Writing down the asymptotes of a rational curve

Answer 13 of 19 · Question 10 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 4 of 5 · find part a.i in this mark scheme extract

	M1	Sets their equation for the normal equal to the curve, makes a **correct** rearrangement to remove any denominator and forms a 3TQ Note this method mark is not dependant.
	A1	Correct exact values for x and y

Type MS · Writing down the asymptotes of a rational curve

Answer 13 of 19 · Question 10 part a.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 5 of 5 · find part a.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing down the asymptotes of a rational curve

Answer 14 of 19 · Question 10 part a.ii · 2 marks

2022 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 15 of 19 · Question 9 part a.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

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Type MS · Writing down the asymptotes of a rational curve

Answer 16 of 19 · Question 10 part b · 2 marks

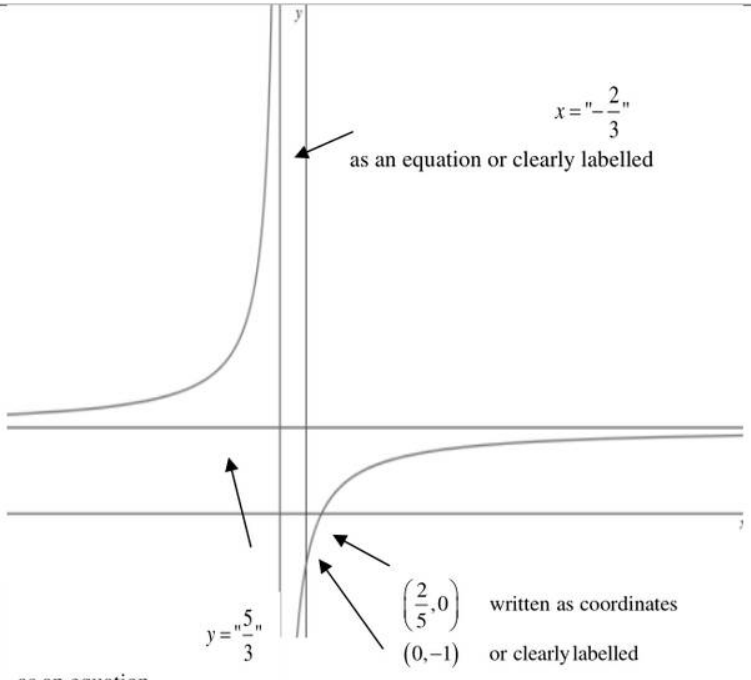
2023 JUN · Paper 02 · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Writing down the asymptotes of a rational curve

Answer 17 of 19 · Question 10 part b.i · 0 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b.i in this mark scheme extract

10 (a)	(i) $\left(\frac{2}{5},0\right)$ (ii) $(0,-1)$	B1 B1 [2]
(b) (i) (ii)	$y=\frac{5}{3}$ $x=-\frac{2}{3}$	B1 B1 [2]
(c)		B1 B1ft B1ft [3]
(d)	$\left\{y=\frac{1}{4}x+\frac{7}{4}\right\}\Rightarrow\frac{1}{4}$	B1
	$\left\{\frac{dy}{dx}=\right\}\frac{5(3x+2)-3(5x-2)}{(3x+2)^2}\left\{=\frac{16}{(3x+2)^2}\right\}$	M1 dM1A1
	$"\frac{1}{4}"="\frac{16}{(3x+2)^2}"\Rightarrow x=...$	M1
	$x=2$ and $y\left\{=\frac{5\times"2"-2}{3\times"2"+2}\right\}=1$	A1
	$y-"1"="-4"(x-"2")$	M1 dM1
	$(0,9)$ and $\left(\frac{9}{4},0\right)$	A1ft
	$\left\{\lvert DE\rvert\right\}=\sqrt{9^2+\left(\frac{9}{4}\right)^2}=\frac{9\sqrt{17}}{4}$	M1A1
		[11]
		Total 18 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 17 of 19 · Question 10 part b.i · 0 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b.i in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	B1	$\left(\frac{2}{5},0\right)$ Accept $x=\frac{2}{5},\{y=0\}$
(a)(ii)	B1	$(0,-1)$ Accept $\{x=0\}, y=-1$, if not labelled, we score by their order
(b)(i)	B1	Correct equation $y=\frac{5}{3}$
(b)(ii)	B1	Correct equation $x=-\frac{2}{3}$
(c)	B1	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (a), at least one section of their curve must pass through one of these intersections.
	B1ft	Two clearly marked asymptotes, ft their (b), labelled as described, there must be at least one section of the curve present, not crossing the curve.
(d)	B1	$\frac{1}{4}$ seen any where in part (d)
	M1	Attempt to differentiate y **Quotient rule**: Numerator must be the difference of two terms (either way round) of the form $P(3x+2)-Q(5x-2),\quad P,Q>0$. Denominator must be correct. **Product rule**: obtains form $P(3x+2)^{-1}-Q(5x-2)(3x+2)^{-2},\ P,Q>0$
	dM1	**Quotient rule**: Either term in the numerator correct **Product rule**: Either term correct Dependent on previous method mark.
	A1	Correct $\frac{dy}{dx}$. May be unsimplified.
	M1	Sets their $\frac{1}{4}$ equal to their $\frac{dy}{dx}$ and attempts to solve for x
	A1	$x=2$ *and* $y=1$
	M1	Finds the gradient of the normal using $\frac{-1}{their\ \frac{1}{4}}$
	dM1	A complete and correct method to find the equation of the normal. If using $y=mx+c$, c must be found. Dependent on previous method mark.
	A1ft	$(0,9)$ and $\left(\frac{9}{4},0\right)$, accept $y=9,\ x=\frac{9}{4}$ Ft their normal equation.
	M1	Correct use of the formula for the length of a line segment, using their values from their points of D and E, may not come from the normal.
	A1	Correct exact length in simplest form

Type MS · Writing down the asymptotes of a rational curve

Answer 17 of 19 · Question 10 part b.i · 0 marks

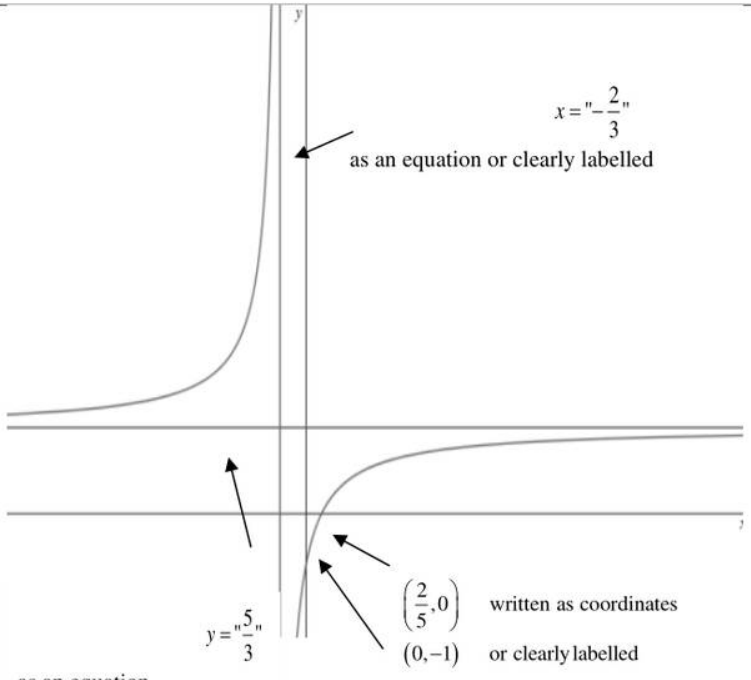
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Type MS · Writing down the asymptotes of a rational curve

Answer 18 of 19 · Question 10 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

10 (a)	(i) $\left(\frac{2}{5},0\right)$ (ii) $(0,-1)$	B1 B1 [2]
(b) (i) (ii)	$y=\frac{5}{3}$ $x=-\frac{2}{3}$	B1 B1 [2]
(c)		B1 B1ft B1ft [3]
(d)	$\left\{y=\frac{1}{4}x+\frac{7}{4}\right\}\Rightarrow\frac{1}{4}$	B1
	$\left\{\frac{dy}{dx}=\right\}\frac{5(3x+2)-3(5x-2)}{(3x+2)^2}\left\{=\frac{16}{(3x+2)^2}\right\}$	M1 dM1A1
	$"\frac{1}{4}"="\frac{16}{(3x+2)^2}"\Rightarrow x=...$	M1
	$x=2$ and $y\left\{=\frac{5\times"2"-2}{3\times"2"+2}\right\}=1$	A1
	$y-"1"="-4"(x-"2")$	M1 dM1
	$(0,9)$ and $\left(\frac{9}{4},0\right)$	A1ft
	$\left\{\lvert DE\rvert\right\}=\sqrt{9^2+\left(\frac{9}{4}\right)^2}=\frac{9\sqrt{17}}{4}$	M1A1
		[11]
		Total 18 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 18 of 19 · Question 10 part b.ii · 2 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	B1	$\left(\frac{2}{5},0\right)$ Accept $x=\frac{2}{5},\{y=0\}$
(a)(ii)	B1	$(0,-1)$ Accept $\{x=0\}, y=-1$, if not labelled, we score by their order
(b)(i)	B1	Correct equation $y=\frac{5}{3}$
(b)(ii)	B1	Correct equation $x=-\frac{2}{3}$
(c)	B1	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (a), at least one section of their curve must pass through one of these intersections.
	B1ft	Two clearly marked asymptotes, ft their (b), labelled as described, there must be at least one section of the curve present, not crossing the curve.
(d)	B1	$\frac{1}{4}$ seen any where in part (d)
	M1	Attempt to differentiate y **Quotient rule**: Numerator must be the difference of two terms (either way round) of the form $P(3x+2)-Q(5x-2),\quad P,Q>0$. Denominator must be correct. **Product rule**: obtains form $P(3x+2)^{-1}-Q(5x-2)(3x+2)^{-2},\ P,Q>0$
	dM1	**Quotient rule**: Either term in the numerator correct **Product rule**: Either term correct Dependent on previous method mark.
	A1	Correct $\frac{dy}{dx}$. May be unsimplified.
	M1	Sets their $\frac{1}{4}$ equal to their $\frac{dy}{dx}$ and attempts to solve for x
	A1	$x=2$ *and* $y=1$
	M1	Finds the gradient of the normal using $\frac{-1}{their\ \frac{1}{4}}$
	dM1	A complete and correct method to find the equation of the normal. If using $y=mx+c$, c must be found. Dependent on previous method mark.
	A1ft	$(0,9)$ and $\left(\frac{9}{4},0\right)$, accept $y=9,\ x=\frac{9}{4}$ Ft their normal equation.
	M1	Correct use of the formula for the length of a line segment, using their values from their points of D and E, may not come from the normal.
	A1	Correct exact length in simplest form

Type MS · Writing down the asymptotes of a rational curve

Answer 18 of 19 · Question 10 part b.ii · 2 marks

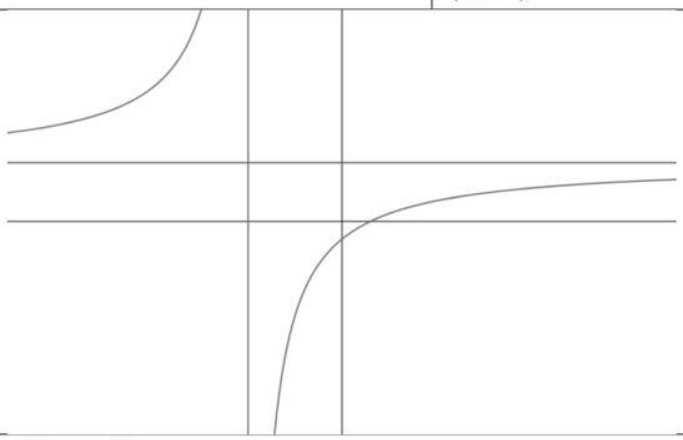
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Type MS · Writing down the asymptotes of a rational curve

Answer 19 of 19 · Question 10 part b · 1 mark

2025 JUN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

10(a) Mark (i) and (ii) together	$-\frac{1}{4}=-\frac{2}{b}\rightarrow b=8$ * $-\frac{4}{3}a+8=0$ oe or $-\frac{b}{a}=-\frac{4}{3}$ $a=6$	B1*cso M1 A1 [3]
(b)	$y=\frac{5}{"6"}$	B1ft [1]
(c)	$\left(\frac{2}{5},0\right)$	B1 [1]
(d)	$\frac{5(6x+8)-6(5x-2)}{(6x+8)^2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion ALT using product rule $(5x-2)(6x+8)^{-1}$ $5(6x+8)^{-1}+(-1)(6)(5x-2)(6x+8)^{-2}$ $\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ $\frac{52}{(6x+8)^2}$ and correct conclusion	M1 dM1 A1 A1 [4]
(e)		B1 (curve) B1ft (asymptotes) B1ft (intersections with axes [3]
(f)	$\frac{13}{100}=\frac{52}{(6x+8)^2}$ oe $(6x+8)^2=400$ **$3x^2+8x-28=0$ oe** $x=2, y=\frac{2}{5}$ $x=-\frac{14}{3}, y=\frac{19}{15}$ oe $\sqrt{\left(("2")-\left("-\frac{14}{3}"\right)\right)^2+\left(\left("\frac{2}{5}"\right)-\left("\frac{19}{15}"\right)\right)^2}$ 6.72	M1 dM1 A1 A1 dM1 A1 [6]
		Total 18 marks

Type MS · Writing down the asymptotes of a rational curve

Answer 19 of 19 · Question 10 part b · 1 mark

2025 JUN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a) (i) and (ii)	B1*cso	For a minimum of the steps shown in the MS to find b, no errors.
	M1	For correct substitution of $x=-\frac{4}{3}$ and $b=8$ into $ax+b=0$
	A1	Correct a.
		Students can complete this in a different order $-\frac{4}{3}a+8=0 \Rightarrow a=6$ $-\frac{b}{a}=-\frac{4}{3}\left(\text{or } -\frac{b}{6}=-\frac{4}{3}\right)\Rightarrow b=8$ B1 M1 A1
(b)	B1ft	Follow through their value of a. $y=\frac{5}{"their 6"}$
(c)	B1	$\left(\frac{2}{5},0\right)$ allow $x=\frac{2}{5}$ $y=0$ $x=\frac{2}{5}$ alone is insufficient
(d)	M1	Attempt the quotient rule. Numerator must be the difference of two terms of the form $A\times(ax+8)-B\times(5x-2)$ or $B\times(5x-2)-A\times(ax+8)$, A and $B>1$. Denominator must be of the form $(ax+8)^2$ $a\neq 0$ Allow follow through of their value of a (general principle of method mark).
	dM1	Either term on the numerator fully correct (either way round), dependent on previous method mark, doesn't need to be simplified. Denominator must be of the form $(ax+8)^2$ Allow follow through of their value of a (general principle of method mark).
	A1	$\frac{52}{(6x+8)^2}$ oe eg $\frac{52}{36x^2+96x+64}$ eg $\frac{13}{(3x+4)^2}$ A simplified expression (denominator expanded acceptable) Dependent on both previous method marks – question states must use calculus, so some method must be shown.
	A1	$\frac{52}{(6x+8)^2}$ and correct conclusion, for example, (the numerator is a positive number) eg the denominator is always positive eg the $(ax+8)^2>0$ and therefore (the fraction) is always positive. No incorrect work for this mark eg don't accept incorrectly simplified gradient.
ALT	M1	Attempt the product rule. $C\times(ax+8)^{-1}+(5x-2)(-(1))(D)\ (ax+8)^{-2}$, C and $D>1$. Allow follow through of their value of a (general principle of method mark).
	M1	Either term correct (either way round), dependent on previous method mark, doesn't need to be simplified
	A1 A1	As main scheme

Type MS · Writing down the asymptotes of a rational curve

Answer 19 of 19 · Question 10 part b · 1 mark

2025 JUN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

(e)	B1	For a negative reciprocal curve drawn anywhere in the grid – there **must** be two branches present, they **must not** cross any asymptotes drawn or implied or cross each other and **must not obviously** 'bend back' on themselves. Mark intention.
	B1ft	Two clearly marked asymptotes $x = -\frac{8}{\text{their } a}$, $y = \frac{5}{\text{their } a}$, ft their value for a, labelled as equations or clearly passing through the correctly labelled numbers on the axis. There must be one section of a negative reciprocal curve present, tending towards these asymptotes. This branch must not obviously cross or bend back from any asymptotes drawn or implied Ignore any other branch/curve/line drawn if there is one branch fulfilling these conditions.
	B1ft	Single curve (or even line) passing through $\left(0, -\frac{1}{4}\right)$ and $\left(\frac{2}{5}, 0\right)$ marked clearly on the graph as coordinates or crossing points on the axis. Ignore any other branch/curve/line present. Allow follow through only of their $(c, 0)$ $c \neq 0$ from part (c)
(f)	M1	Sets their differentiated function from part (d) = $\frac{13}{100}$ Allow any function = $\frac{13}{100}$ coming from working in part (d) or re-started work to find the derivative. The function may not be the original function.
	dM1	For correctly rearranging an equation of the form $\frac{13}{100} = \frac{Q}{(cx+d)^2}$ to $(cx+d)^2 = R$ using their a, or to the form $px^2 + qx + r = 0$ $p, q, r \neq 0$ with 2 out of 3 of p, q, r correct. leading to $x =$
	A1	For $x = 2, y = \frac{2}{5}$ or $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	A1	For $x = 2, y = \frac{2}{5}$ and $x = -\frac{14}{3}, y = \frac{19}{15}$ The pairing must be clear (may be implied in further working)
	dM1	For correct use of the formula to find the length of a line, using their values for x and y. Dependent on **first** method mark only.
	A1	For awrt 6.72

Type MS · Writing down the asymptotes of a rational curve

Answer 19 of 19 · Question 10 part b · 1 mark

2025 JUN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks